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Simplifying algebraic expressions
Simplifying algebraic expressions involves manipulating and rewriting algebraic terms in a simpler or more compact form without changing their original value. This can apply to linear or non-linear expressions.

Example 1: Simplify the expression $(x + y) + (3x - y)$.

To simplify the expression $(x + y) + (3x - y)$, you can combine like terms.
i) Remove parentheses:
The expression $(x + y)$ has no operation between its terms, so it can be rewritten as $x + y$. Similarly, $(3x - y)$ becomes $3x - y$. The combined expression is:

$$x + y + 3x - y$$

ii) Combine like terms: - Group similar terms and use cancellations as needed. Adding and subtracting like terms gives:

$$x + \cancel{y} + 3x - \cancel{y}$$

Now, combine terms with the same variable to get the simplified expression:

$$4x$$

Thus, the simplified expression for $(x + y) + (3x - y)$ is: $4x$.

Example 2: Simplify $(x^2 + 3x + 5) + (x - 2)^2$

To simplify the expression $(x^2 + 3x + 5) + (x - 2)^2$, we first expand $(x - 2)^2$ and then combine like terms.
i) Expand $(x - 2)^2$:

$$(x - 2) \times (x - 2) = x^2 - 2x - 2x + 4 = x^2 - 4x + 4$$

ii) Combine the expanded form with the first part of the expression:

$$(x^2 + 3x + 5) + (x^2 - 4x + 4)$$

iii) Add and combine like terms:

- Combine the x^2 terms: $x^2 + x^2 = 2x^2$.

- Combine the x terms: $3x - 4x = -x$.
- Combine the constants: $5 + 4 = 9$.

iv) Rewrite the simplified expression:

$$2x^2 - x + 9$$

Thus, the simplified expression for $(x^2 + 3x + 5) + (x - 2)^2$ is: $2x^2 - x + 9$.